$$a_{n+1} = \frac{\left(n+\frac{1}{2}\right)2n-3}{\left(n+\frac{2}{2}\right)(2n+2)} a_n, \qquad n \geq 0$$

Putting $n = 0, 1, 2, \ldots$

$$a_1 = \frac{-3}{\frac{3}{2}\cdot 2} a_0 = -a_0$$

$$a_2 = \frac{\frac{3}{2}\cdot 2 - 3}{\frac{5}{2}\cdot 4} = 0$$

Since $a_2 = 0, a_3 = a_4 = a_5 = \ldots = 0$

Substituting $r = \frac{1}{2}$ and values of $a_1, a_2, \ldots$ in Eq. (2),

$$y_2 = x^{\frac{1}{2}}(a_0 - a_0 x + 0)$$
$$= a_0\sqrt{x}(1-x)$$

Hence, the general solution is

$$y = c_1 y_1 + c_2 y_2$$
$$= c_1 a_0 (1 - 3x + x^2 - \cdots) + c_2 a_0 \sqrt{x}(1-x)$$
$$= A(1 - 3x + x^2 - \cdots) + B\sqrt{x}(1-x)$$

where $A = c_1 a_0$, $B = c_2 a_0$

Example 3

Find the series solution of the equation $xy'' + y' - y = 0$ *about* $x_0 = 0$

Solution

$$xy'' + y' - y = 0 \qquad \ldots(1)$$

$P_0(x) = x = 0$ at $x = 0$

Hence, $x = 0$ is a singular point.

$$y'' + \frac{1}{x}y' - \frac{1}{x}y = 0$$

$$P(x) = \frac{1}{x}, \qquad Q(x) = -\frac{1}{x}$$

Since $xP(x) = 1$ and $x^2Q(x) = -x$ are analytic (i.e., differentiable) at $x = 0$, it is a regular singular point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^{n+r} = x^r (a_0 + a_1 x + a_2 x^2 + \cdots) \quad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r-2}$$

Substituting in Eq. (1),

$$x\sum_{n=0}^{\infty} (n+r)(n+r-1)a_n x^{n+r-2} + \sum_{n=0}^{\infty} (n+r)a_n x^{n+r-1} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} (n+r)(n+r-1)a_n x^{n+r-1} + \sum_{n=0}^{\infty} (n+r)a_n x^{n+r-1} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} (n+r)(n+r-1+1)a_n x^{n+r-1} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} (n+r)^2 a_n x^{n+r-1} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

To obtain the common power of x in each term, putting $n = m + 1$ in the first term, we get

$$\sum_{m=-1}^{\infty} (m+1+r)^2 a_{m+1} x^{m+r} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=-1}^{\infty} (n+r+1)^2 a_{n+1} x^{n+r} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$a_0 r^2 x^{-1+r} + \sum_{n=0}^{\infty} (n+r+1)^2 a_{n+1} x^{n+r} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

Equating the coefficient of the lowest degree term to zero, the indicial equation is

$$a_0 r^2 = 0$$

$$r = 0, 0 \quad [\because \quad a_0 \neq 0]$$

which is a double root.

Equating the coefficient of x^{n+r} to zero.

$$a_{n+1}(n+r+1)^2 - a_n = 0, \qquad n \geq 0$$

$$a_{n+1} = \frac{1}{(n+r+1)^2} a_n, \qquad n \geq 0$$

Putting $n = 0, 1, 2, \ldots$

$$a_1 = \frac{1}{(r+1)^2} a_0$$

$$a_2 = \frac{1}{(r+2)^2} a_1 = \frac{1}{(r+2)^2 (r+1)^2} a_0$$

and so on.

Substituting in Eq. (2),

$$y = x^r \left[a_0 + \frac{1}{(r+1)^2} a_0 x + \frac{1}{(r+1)^2 (r+2)^2} a_0 x^2 + \cdots \right]$$

$$= a_0 x^r \left[1 + \frac{1}{(r+1)^2} x + \frac{1}{(r+1)^2 (r+2)^2} x^2 + \cdots \right] \quad \ldots(3)$$

For the first solution, putting $r = 0$ in Eq. (3),

$$y_1 = (y)_{r=0} = a_0 \left(1 + x + \frac{x^2}{4} + \cdots \right)$$

For the second solution, differentiating Eq. (3) w.r.t. r,

$$\frac{\partial y}{\partial r} = a_0 x^r \log x \left[1 + \frac{1}{(r+1)^2} x + \frac{1}{(r+1)^2 (r+2)^2} x^2 + \cdots \right]$$

$$+ a_0 x^r \left[-\frac{2}{(r+1)^3} x - \frac{2}{(r+1)^3} \cdot \frac{x^2}{(r+2)^2} - \frac{2}{(r+2)^3} \cdot \frac{x^2}{(r+1)^2} + \cdots \right]$$

Putting $r = 0$,

$$y_2 = \left(\frac{\partial y}{\partial r} \right)_{r=0}$$

$$= a_0 \log x \left(1 + x + \frac{x^2}{4} + \cdots \right) - 2a_0 \left(x + \frac{x^2}{4} + \frac{x^2}{8} + \cdots \right)$$

$$= a_0 \log x \left(1 + x + \frac{x^2}{4} + \cdots \right) - 2a_0 \left(x + \frac{3}{8} x^2 + \cdots \right)$$

Hence, the general solution is

$$y = c_1 y_1 + c_2 y_2$$

$$= c_1 a_0 \left(1 + x + \frac{x^2}{4} + \cdots \right) + c_2 a_0 \log x \left(1 + x + \frac{x^2}{4} + \cdots \right) - 2c_2 a_0 \left(x + \frac{3}{8} x^2 + \cdots \right)$$

$$= (A + B \log x) \left(1 + x + \frac{x^2}{4} + \cdots \right) - 2B \left(x + \frac{3}{8} x^2 + \cdots \right)$$

where $A = c_1 a_0$, $B = c_2 a_0$

Example 4

Solve the differential equation by Frobenius method

$$x(x-1)y'' + (3x-1)y' + y = 0 \quad \text{at } x = 0$$

Solution

$$x(x-1)y'' + (3x-1)y' + y = 0 \qquad \ldots(1)$$

$$P_0(x) = x(x-1) = 0 \qquad \text{at } x = 0$$

Hence, $x = 0$ is a singular point.

$$y'' + \frac{(3x-1)}{x(x-1)}y' + \frac{1}{x(x-1)}y = 0$$

$$P(x) = \frac{3x-1}{x(x-1)}, \qquad Q(x) = \frac{1}{x(x-1)}$$

Since $xP(x) = \dfrac{3x-1}{x-1}$ and $x^2Q(x) = \dfrac{x}{x-1}$ are analytic (i.e., differentiable) at $x = 0$, it is a regular singular point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^{n+r} = x^r(a_0 + a_1x + a_2x^2 + \cdots) \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n(n+r)x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n(n+r)(n+r-1)x^{n+r-2}$$

Substituting in Eq. (1),

$$(x^2 - x)\sum_{n=0}^{\infty}(n+r)(n+r-1)a_n x^{n+r-2} + (3x-1)\sum_{n=0}^{\infty}(n+r)a_n x^{n+r-1} + \sum_{n=0}^{\infty}a_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty}(n+r)(n+r-1)a_n x^{n+r} - \sum_{n=0}^{\infty}(n+r)(n+r-1)a_n x^{n+r-1} + 3\sum_{n=0}^{\infty}(n+r)a_n x^{n+r}$$

$$- \sum_{n=0}^{\infty}(n+r)a_n x^{n+r-1} + \sum_{n=0}^{\infty}a_n x^{n+r} = 0$$

$$-\sum_{n=0}^{\infty}(n+r)(n+r-1+1)a_n x^{n+r-1} + \sum_{n=0}^{\infty}\left[(n+r)(n+r-1) + 3(n+r) + 1\right]a_n x^{n+r} = 0$$

To obtain a common power of x in each term, putting $n = m + 1$ in the first term, we get

$$-\sum_{m=-1}^{\infty}(m+1+r)^2 a_{m+1}x^{m+r} + \sum_{n=0}^{\infty}\left[(n+r)(n+r+2)+1\right]a_n x^{n+r} = 0$$

Since m is a dummy variable, replacing m by n,

$$-\sum_{n=-1}^{\infty}(n+r+1)^2 a_{n+1}x^{n+r}+\sum_{n=0}^{\infty}\left[(n+r)(n+r+2)+1\right]a_n x^{n+r}=0$$

$$-r^2 a_0 x^{-1+r}-\sum_{n=0}^{\infty}(n+r+1)^2 a_{n+1}x^{n+r}+\sum_{n=0}^{\infty}\left[(n+r)(n+r+2)+1\right]a_n x^{n+r}=0$$

Equating the coefficient of the lowest degree term (i.e., x^{r-1}) to zero,

$$-r^2 a_0 = 0$$

$$r = 0, 0 \quad [\because\ a_0 \neq 0]$$

which is a double root.

Equating the coefficient of x^{n+r} to zero,

$$-(n+r+1)^2 a_{n+1}+\left[(n+r)(n+r+2)+1\right]a_n = 0, \qquad n \geq 0$$

$$a_{n+1} = \frac{(n+r)(n+r+2)+1}{(n+r+1)^2}a_n, \qquad n \geq 0$$

$$= \frac{(n+r)^2+2(n+r)+1}{(n+r+1)^2}a_n, \qquad n \geq 0$$

$$= \frac{(n+r+1)^2}{(n+r+1)^2}a_n, \qquad n \geq 0$$

$$= a_n$$

Putting $n = 0, 1, 2, \ldots$

$$a_1 = a_0$$

$$a_2 = a_1 = a_0$$

and so on.

Substituting in Eq. (2),

$$y = x^r(a_0 + a_0 x + a_0 x^2 + \cdots)$$

$$= a_0 x^r(1 + x + x^2 + \cdots) \qquad \ldots(3)$$

For the first solution, putting $r = 0$ in Eq. (3),

$$y_1 = (y)_{r=0} = a_0(1 + x + x^2 + \ldots)$$

For the second solution, differentiating Eq. (3) w.r.t. r,

$$\frac{\partial y}{\partial r} = a_0 x^r \log x(1 + x + x^2 \cdots)$$

Putting $r = 0$,

$$y_2 = \left(\frac{\partial y}{\partial r}\right)_{r=0}$$

$$= a_0 \log x(1 + x + x^2 + \cdots)$$

Hence, the general solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 a_0 (1 + x + x^2 \cdots) + c_2 a_0 \log x (1 + x + x^2 + \cdots) \\ &= (A + B \log x)(1 + x + x^2 + \cdots) \\ &= (A + B \log x)(1 - x)^{-1} \end{aligned}$$

where $A = c_1 a_0$, $B = c_2 a_0$

Example 5

Obtain the series solution of the differential equation

$$xy'' + y' + xy = 0$$

Solution

$$xy'' + y' + xy = 0 \qquad \ldots(1)$$

$P_0(x) = x = 0$ at $x = 0$

Hence, $x = 0$ is a singular point.

$$y'' + \frac{1}{x} y' + y = 0$$

$$P(x) = \frac{1}{x}, \qquad Q(x) = 1$$

Since $xP(x) = 1$ and $x^2 Q(x) = x^2$ are analytic (i.e., differentiable) at $x = 0$, it is a regular singular point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^{n+r} = x^r (a_0 + a_1 x + a_2 x^2 + \cdots) \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting in Eq. (1)

$$x \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} + \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} + x \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-1} + \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

$$\sum_{n=0}^{\infty} (n+r-1+1) a_n x^{n+r-1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

$$\sum_{n=0}^{\infty}(n+r)^2 a_n x^{n+r-1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

To obtain a common power of x in each term, putting $n = m + 2$ in the first term, we get

$$\sum_{m=-2}^{\infty}(m+2+r)^2 a_{m+2} x^{m+r+1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=-2}^{\infty}(n+r+2)^2 a_{n+2} x^{n+r+1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

$$r^2 a_0 x^{r-1} + (r+1)^2 a_1 x^r + \sum_{n=0}^{\infty}(n+r+2)^2 a_{n+2} x^{n+r+1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

Equating the coefficient of the lowest degree term to zero, the indicial equation is

$$a_0 r^2 = 0$$

$$r = 0, 0 \quad [\because \quad a_0 \neq 0]$$

which is a double root.

Equating the coefficient of x^r and x^{n+r+1} to zero,

$$a_1(r+1)^2 = 0$$

$$a_1 = 0 \quad [\because \quad r \neq -1]$$

and $a_{n+2}(n+r+2)^2 + a_n = 0, \qquad n \geq 0$

$$a_{n+2} = -\frac{1}{(n+r+2)^2} a_n, \quad n \geq 0$$

Putting $n = 0, 1, 2, \ldots$

$$a_2 = -\frac{1}{(r+2)^2} a_0$$

$$a_3 = -\frac{1}{(r+3)^2} a_1 = 0$$

$$a_4 = -\frac{1}{(r+4)^2} a_2 = -\frac{1}{(r+4)^2}\left[-\frac{1}{(r+2)^2} a_0\right] = \frac{1}{(r+2)^2(r+4)^2} a_0$$

and so on.

Substituting in Eq. (2),

$$y = x^r\left[a_0 + 0 \cdot x - \frac{1}{(r+2)^2} a_0 x^2 + 0 \cdot x^3 + \frac{1}{(r+2)^2(r+4)^2} a_0 x^4 + \cdots\right]$$

$$= a_0 x^r\left[1 - \frac{1}{(r+2)^2} x^2 + \frac{1}{(r+2)^2(r+4)^2} x^4 - \cdots\right] \qquad \ldots(3)$$

For the first solution, putting $r = 0$ in Eq. (3),

$$y_1 = (y)_{r=0}$$
$$= a_0\left(1-\frac{x^2}{2^2}+\frac{x^4}{2^2\cdot 4^2}-\cdots\right)$$

For the second solution, differentiating Eq. (3) w.r.t. r,

$$\frac{\partial y}{\partial r} = a_0 x^r \log x\left[1-\frac{1}{(r+2)^2}x^2+\frac{1}{(r+2)^2(x+4)^2}x^4-\cdots\right]$$
$$+ a_0 x^r\left[1+\frac{2}{(r+2)^3}x^2-\frac{2}{(r+2)^3(r+4)^2}x^4-\frac{2}{(r+2)^2(r+4)^3}x^4+\ldots\right]$$

Putting $r = 0$,

$$y_2 = \left(\frac{\partial y}{\partial r}\right)_{r=0}$$
$$= a_0 \log x\left(1-\frac{x^2}{2^2}+\frac{x^4}{2^2\cdot 4^2}-\cdots\right)+a_0\left(1+\frac{x^2}{2^2}-\frac{x^4}{2^2\cdot 4^2}-\frac{x^4}{2\cdot 4^3}+\cdots\right)$$
$$= a_0 \log x\left(1-\frac{x^2}{2^2}+\frac{x^4}{2^2\cdot 4^2}-\cdots\right)+a_0\left(1+\frac{x^2}{2^2}-\frac{3}{2^3\cdot 4^2}x^4+\cdots\right)$$

Hence, the general solution is

$$y = c_1y_1 + c_2y_2$$
$$= c_1a_0\left(1-\frac{x^2}{2^2}+\frac{x^4}{2^2\cdot 4^2}-\cdots\right)+c_2a_0\log x\left(1-\frac{x^2}{2^2}+\frac{x^4}{2^2\cdot 4^2}-\cdots\right)$$
$$+ c_2a_0\left(1+\frac{x^2}{2^2}-\frac{3}{2^3\cdot 4^2}x^4+\cdots\right)$$
$$= (A + B\log x)\left(1-\frac{x^2}{2^2}+\frac{x^4}{2^2\cdot 4^2}-\cdots\right)+B\left(1+\frac{x^2}{2^2}-\frac{3}{2^3\cdot 4^2}x^4+\cdots\right)$$

where $A = c_1 a_0$, $B = c_2 a_0$

Example 6

Find a series solution of the differential equation

$x^2y'' + x^3y' + (x^2 - 2)y = 0$ *about* $x = 0$.

Solution

$$x^2y'' + x^3y' + (x^2 - 2)y = 0 \quad \ldots(1)$$

$P_0(x) = x^2 = 0$ at $x = 0$

Hence, $x = 0$ is a singular point.

$$y'' + xy' + \frac{x^2 - 2}{x^2} y = 0$$

$$P(x) = x, \qquad Q(x) = \frac{x^2 - 2}{x^2}$$

Since $xP(x) = x^2$ and $x^2Q(x) = x^2 - 2$ are analytic (i.e., differentiable) at $x = 0$, it is a regular singular point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^{n+r} = x^r (a_0 + a_1 x + a_2 x^2 + \cdots) \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r-2}$$

Substituting in Eq. (1)

$$x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} + x^3 \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} + (x^2 - 2) \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r} + \sum_{n=0}^{\infty} (n+r) a_n x^{n+r+2} + \sum_{n=0}^{\infty} a_n x^{n+r+2} - 2 \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} \left[(n+r)(n+r-1) - 2\right] a_n x^{n+r} + \sum_{n=0}^{\infty} (n+r+1) a_n x^{n+r+2} = 0$$

To obtain a common power of x in each term, putting $n = m - 2$ in the second term, we get

$$\sum_{n=0}^{\infty} \left[(n+r)(n+r-1) - 2\right] a_n x^{n+r} + \sum_{m=2}^{\infty} (m+r-1) a_{m-2} x^{m+r} = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=0}^{\infty} \left[(n+r)(n+r-1) - 2\right] a_n x^{n+r} + \sum_{n=2}^{\infty} (n+r-1) a_{n-2} x^{n+r} = 0$$

$$\left[r(r-1) - 2\right] a_0 x^r + \left[(1+r)r - 2\right] a_1 x^{r+1} + \sum_{n=2} \left[(n+r)(n+r-1) - 2\right] a_n x^{n+r}$$

$$+ \sum_{n=2}^{\infty} (n+r-1) a_{n-2} x^{n+r} = 0$$

Equating the coefficient of the lowest degree term to zero, the indicial equation is

$$a_0[r(r-1)-2]=0$$

$$a_0(r+1)(r-2)=0$$

$$r=-1,\ r=2 \qquad [\because\ a_0 \neq 0]$$

Equating the coefficient of x^{r+1} to zero,

$$[r(r+1)-2]a_1=0$$

$$(r-1)(r+2)a_1=0$$

$$a_1=0 \qquad [\because\ r\neq 1,\ r\neq -2]$$

Equating the coefficient of x^{n+r} to zero,

$$[(n+r)(n+r-1)-2]a_n+(n+r-1)a_{n-2}=0, \qquad n\geq 2$$

$$a_n=-\frac{(n+r-1)}{(n+r-1)(n+r)-2}a_{n-2}, \qquad n\geq 2 \qquad \ldots(3)$$

For the first solution, putting $r=-1$ in Eq. (3),

$$a_n=-\frac{(n-2)}{(n-2)(n-1)-2}a_{n-2}, \qquad n\geq 2$$

$$=-\frac{(n-2)}{(n^2-3n+2-2)}a_{n-2}, \qquad n\geq 2$$

$$=-\frac{(n-2)}{n(n-3)}a_{n-2}, \qquad n\geq 2$$

Putting $n=2, 3, 4, \ldots$

$$a_2=0$$

$$a_3=\lim_{n\to 3}\left[-\frac{(n-2)}{n(n-3)}a_1\right] \qquad \left[\frac{0}{0}\text{form},\ \because\ a_1=0\right]$$

$$=\lim_{n\to 3}\left[-\frac{a_1}{2n-3}\right] \qquad \text{[Using L'Hospital's rule]}$$

$$=-\frac{a_1}{3}=0 \qquad [\because\ a_1=0]$$

$$a_4=-\frac{2}{4\cdot 1}a_2=0$$

$$a_5=-\frac{3}{5\cdot 2}a_3=0$$

and so on.

Putting $r = -1$ and values of $a_1, a_2, a_3, \ldots$ in Eq. (2),

$$y_1 = (y)_{r=-1}$$
$$= x^{-1}(a_0 + 0 \cdot x + 0 \cdot x^2 + \cdots)$$
$$= \frac{a_0}{x}$$

For the second solution, putting $r = 2$ in Eq. (3),

$$a_n = -\frac{(n+1)}{(n+1)(n+2)-2} a_{n-2}, \quad n \geq 2$$
$$= -\frac{(n+1)}{(n^2+3n)} a_{n-2}, \quad n \geq 2$$
$$= -\frac{n+1}{n(n+3)} a_{n-2}, \quad n \geq 2$$

Putting $n = 2, 3, 4, \ldots$

$$a_2 = -\frac{3}{2 \cdot 5} a_0 = -\frac{3}{10} a_0$$
$$a_3 = -\frac{4}{3 \cdot 6} a_1 = 0 \qquad [\because\ a_1 = 0]$$
$$a_4 = -\frac{5}{4 \cdot 7} a_2 = -\frac{5}{28}\left(-\frac{3}{10} a_0\right) = \frac{3}{56} a_0$$

and so on.

Putting $r = 2$ and values of $a_1, a_2, a_3, \ldots$ in Eq. (2),

$$y_2 = (y)_{r=2}$$
$$= x^2\left(a_0 + 0 \cdot x - \frac{3}{10} a_0 x^2 + 0 \cdot x^3 + \frac{3}{56} a_0 x^4 + \cdots\right)$$
$$= a_0 x^2\left(1 - \frac{3}{10} x^2 + \frac{3}{56} x^4 - \cdots\right)$$

Hence, the general solution is

$$y = c_1 y_1 + c_2 y_2$$
$$= c_1 \frac{a_0}{x} + c_2 a_0 x^2\left(1 - \frac{3}{10} x^2 + \frac{3}{56} x^4 - \cdots\right)$$
$$= \frac{A}{x} + B x^2\left(1 - \frac{3}{10} x^2 + \frac{3}{56} x^4 - \cdots\right)$$

where $A = c_1 a_0$, $B = c_2 a_0$

Example 7

Find the series solution of the differential equation

$$(x^2 - x)y'' - xy' + y = 0 \text{ about } x = 0.$$

Solution

$$(x^2 - x)y'' - xy' + y = 0 \qquad \ldots(1)$$

$$P_0(x) = x^2 - x = 0 \text{ at } x = 0$$

Hence, $x = 0$ is a singular point.

$$y'' - \frac{1}{x-1}y' + \frac{1}{x(x-1)}y = 0$$

$$P(x) = -\frac{1}{x-1}, \quad Q(x) = \frac{1}{x(x-1)}$$

Since $xP(x) = -\dfrac{x}{x-1}$ and $x^2Q(x) = \dfrac{x}{x-1}$ are analytic (i.e., differentiable) at $x = 0$, it is a regular singular point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^{n+r} = x^r(a_0 + a_1x + a_2x^2 + \cdots) \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n(n+r)x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n(n+r)(n+r-1)x^{n+r-2}$$

Substituting in Eq. (1),

$$(x^2 - x)\sum_{n=0}^{\infty}(n+r)(n+r-1)a_nx^{n+r-2} - x\sum_{n=0}^{\infty}(n+r)a_nx^{n+r-1} + \sum_{n=0}^{\infty}a_nx^{n+r} = 0$$

$$\sum_{n=0}^{\infty}(n+r)(n+r-1)a_nx^{n+r} - \sum_{n=0}^{\infty}(n+r)(n+r-1)a_nx^{n+r-1} - \sum_{n=0}^{\infty}(n+r)a_nx^{n+r}$$

$$+ \sum_{n=0}^{\infty}a_nx^{n+r} = 0$$

$$\sum_{n=0}^{\infty}\left[(n+r)(n+r-1) - (n+r) + 1\right]a_nx^{n+r} - \sum_{n=0}^{\infty}(n+r)(n+r-1)a_nx^{n+r-1} = 0$$

$$-\sum_{n=0}^{\infty}(n+r)(n+r-1)a_nx^{n+r-1} + \sum_{n=0}^{\infty}\left[(n+r)(n+r-2)+1\right]a_nx^{n+r} = 0$$

To obtain a common power of x in each term, putting $n = m + 1$ in the first term, we get

$$-\sum_{m=-1}^{\infty}(m+1+r)(m+r)a_{m+1}x^{m+r}+\sum_{n=0}^{\infty}\left[(n+r)(n+r-2)+1\right]a_n x^{n+r}=0$$

Since m is a dummy variable, replacing m by n and multiplying by the negative sign,

$$\sum_{n=-1}^{\infty}(n+r+1)(n+r)a_{n+1}x^{n+r}-\sum_{n=0}^{\infty}\left[(n+r)(n+r-2)+1\right]a_n x^{n+r}=0$$

$$r(r-1)a_0x^{r-1}+\sum_{n=0}^{\infty}(n+r+1)(n+r)a_{n+1}x^{n+r}-\sum_{n=0}^{\infty}\left[(n+r)(n+r-2)+1\right]a_n x^{n+r}=0$$

Equating the coefficient of the lowest degree term to zero, the indicial equation is

$$r(r-1)\,a_0=0$$

$$r=0,\quad r=1 \qquad [\because\ a_0\neq 0]$$

Equating the coefficient of x^{n+r} to zero,

$$(n+r+1)(n+r)a_{n+1}-\left[(n+r)(n+r-2)+1\right]a_n=0,\quad n\geq 0$$

$$a_{n+1}=\frac{(n+r)(n+r-2)+1}{(n+r)(n+r+1)}a_n,\qquad n\geq 0$$

$$=\frac{(n+r)^2-2(n+r)+1}{(n+r)(n+r+1)}a_n,\qquad n\geq 0$$

$$=\frac{(n+r-1)^2}{(n+r)(n+r+1)}a_n,\qquad n\geq 0$$

Putting $n = 0, 1, 2, \ldots$

$$a_1=\frac{(r-1)^2}{r(r+1)}a_0$$

$$a_2=\frac{r^2}{(r+1)(r+2)}a_1$$

and so on.

At $r = 0$, $a_1 = \infty$ and, hence, $a_2 = a_3 = \ldots = \infty$

To obtain the solution of Eq. (1), the procedure is modified by assuming $a_0 = c_0 r$, $c_0 \neq 0$.

Substituting $a_0 = c_0 r$ in $a_1, a_2, \ldots$

$$a_1=\frac{(r-1)^2}{r(r+1)}c_0 r=\frac{(r-1)^2}{(r+1)}c_0$$

$$a_2=\frac{r^2}{(r+1)(r+2)}\frac{(r-1)^2}{(r+1)}c_0=\frac{r^2(r-1)^2}{(r+1)^2(r+2)}c_0$$

and so on.

Substituting in Eq. (2),

$$y = c_0 x^r \left[r + \frac{(r-1)^2}{(r+1)} x + \frac{r^2 (r-1)^2}{(r+1)^2 (r+2)} x^2 + \cdots \right] \quad \ldots(3)$$

For the first solution, putting $r = 0$ in Eq. (3),

$$y_1(y)_{r=0} = c_0(0 + x + 0) = c_0 x$$

For the second solution, putting $r = 1$ in Eq. (3),

$$y_2 = (y)_{r=1} = c_0 x = y_1$$

Hence, differentiating Eq. (3) w.r.t. r to obtain the second solution,

$$\frac{\partial y}{\partial r} = c_0 x^r \log x \left[r + \frac{(r-1)^2}{(r+1)} x + \begin{pmatrix} \text{(terms containing higher powers of} \\ r \text{ in the numerator)} \end{pmatrix} \right]$$

$$+ c_0 x^r \left[1 + \frac{2(r-1)}{r+1} x - \frac{(r-1)^2}{(r+1)^2} x + \begin{pmatrix} \text{(terms containing higher powers of} \\ r \text{ in the numerator)} \end{pmatrix} \right]$$

Putting $r = 0$,

$$y_2 = \left(\frac{\partial y}{\partial r} \right)_{r=0} = c_0 \log x(0 + x + 0) + c_0(1 - 2x - x + 0)$$

$$= c_0(x \log x + 1 - 3x)$$

Hence, the general solution is

$$y = c_1 y_1 + c_2 y_2$$
$$= c_1 c_0 x + c_2 c_0 (x \log x + 1 - 3x)$$
$$= Ax + B(x \log x + 1 - 3x)$$
$$= (A + B \log x)x + B(1 - 3x)$$

where $A = c_1 c_0$, $B = c_2 c_0$

Example 8

Solve in series the differential equation

$$xy'' + 2y' + xy = 0$$

Solution

$$xy'' + 2y' + xy = 0 \quad \ldots(1)$$

$P_0(x) = x = 0$ at $x = 0$

Hence, $x = 0$ is a singular point.

$$y'' + \frac{2}{x} y' + y = 0$$

$$P(x) = \frac{2}{x}, \qquad Q(x) = 1$$

Since $xP(x) = 2$ and $x^2Q(x) = x^2$ are analytic (i.e., differentiable) at $x = 0$, it is a regular singular point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^{n+r} = x^r (a_0 + a_1 x + a_2 x^2 + \cdots) \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting in Eq. (1),

$$x \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} + 2 \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} + x \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-1} + 2 \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

$$\sum_{n=0}^{\infty} (n+r)(n+r-1+2) a_n x^{n+r-1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

$$\sum_{n=0}^{\infty} (n+r)(n+r+1) a_n x^{n+r-1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

To obtain a common power of x in each term, putting $n = m + 2$ in the first term, we get

$$\sum_{m=-2}^{\infty} (m+2+r)(m+2+r+1) a_{m+2} x^{m+r+1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=-2}^{\infty} (n+r+2)(n+r+3) a_{n+2} x^{n+r+1} + \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

$$r(r+1) a_0 x^{r-1} + (r+1)(r+2) a_1 x^r + \sum_{n=0}^{\infty} (n+r+2)(n+r+3) a_{n+2} x^{n+r+1}$$

$$+ \sum_{n=0}^{\infty} a_n x^{n+r+1} = 0$$

Equating the coefficient of the lowest degree term to zero, the indicial equation is

$$a_0 r(r+1) = 0$$

$$r = 0, \quad r = -1 \qquad [\because \quad a_0 \neq 0]$$

Equating the coefficient of x^r to zero,

$$a_1 (r+1)(r+2) = 0$$

$$a_1 \neq 0 \qquad [\because \ r = -1]$$

Thus, for the smaller root $r = -1$, a_1 is an arbitrary constant.

For $r = 0$, $a_1 = 0$.

Equating the coefficient of x^{n+r+1} to zero,

$$(n+r+2)(n+r+3)a_{n+2} + a_n = 0, \qquad n \geq 0$$

$$a_{n+2} = -\frac{1}{(n+r+2)(n+r+3)} a_n, \qquad n \geq 0 \qquad \text{...(3)}$$

For the first solution, putting $r = -1$,

$$a_{n+2} = -\frac{1}{(n+1)(n+2)} a_n, \qquad n \geq 0$$

Putting $n = 0, 1, 2, \ldots$

$$a_2 = -\frac{1}{2} a_0 = -\frac{1}{2!} a_0$$

$$a_3 = -\frac{1}{2 \cdot 3} a_1 = -\frac{1}{3!} a_1$$

$$a_4 = -\frac{1}{3 \cdot 4} a_2 = -\frac{1}{3 \cdot 4}\left(-\frac{1}{2!} a_0\right) = \frac{1}{4!} a_0$$

$$a_5 = -\frac{1}{4 \cdot 5} a_3 = -\frac{1}{4 \cdot 5}\left(-\frac{1}{3!} a_1\right) = \frac{1}{5!} a_1$$

and so on.

Substituting $r = -1$ and values of $a_1, a_2, a_3, \ldots$ in Eq. (2),

$$y_1 = (y)_{r=-1}$$

$$= x^{-1}\left(a_0 + a_1 x - \frac{1}{2!} a_0 x^2 - \frac{1}{3!} a_1 x^3 + \frac{1}{4!} a_0 x^4 + \frac{1}{5!} a_1 x^5 - \cdots\right)$$

$$= \frac{a_0}{x}\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) + a_1\left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \cdots\right)$$

For the second solution, putting $r = 0$ in Eq. (3),

$$a_{n+2} = -\frac{1}{(n+2)(n+3)} a_n, \qquad n \geq 0$$

Putting $n = 0, 1, 2, \ldots$

$$a_2 = -\frac{1}{2 \cdot 3} a_0 = -\frac{1}{3!} a_0$$

$$a_3 = -\frac{1}{3 \cdot 4} a_1 = 0 \quad [\because \ a_1 = 0]$$

$$a_4 = -\frac{1}{4 \cdot 5} a_2 = -\frac{1}{4 \cdot 5}\left(-\frac{1}{3!} a_0\right) = \frac{1}{5!} a_0$$

$$a_5 = -\frac{1}{5 \cdot 6} a_3 = 0$$

and so on.

Substituting $r = 0$ and values of $a_1, a_2, a_3, \ldots$ in Eq. (2),

$$y_2 = (y)_{r=0}$$

$$= x^0\left(a_0 + 0 \cdot x - \frac{1}{3!} a_0 x^2 + 0 \cdot x^3 + \frac{1}{5!} a_0 x^4 + 0 \cdot x^5 - \cdots\right)$$

$$= a_0\left(1 - \frac{x^2}{3!} - \frac{x^4}{5!} - \cdots\right)$$

But this solution is a constant multiple of the second solution in Eq. (4).

Hence, the solution represented by Eq. (4) is the required general solution with two arbitrary constants as a_0 and a_1.